

20. The Idea of Renormalization.

$$H = \underbrace{H_0}_{\text{free Hamiltonian}} + \underbrace{g H_I}_{\text{interaction Hamiltonian}} \rightarrow \text{coupling parameter}$$

S : scattering operator

$$P = (p^0, p^1, p^2, p^3). \quad p_1, p_2, p_3, p_4 : \text{distinct.}$$

$$\begin{aligned} & \langle P_3, P_4 | S | P_1, P_2 \rangle \\ &= \left(\underbrace{g - ig^2 M + O(g^3)}_{\text{unknown}} \right) \cdot \frac{-i(2\pi)^4 \delta^4(p_3 + p_4 - p_1 - p_2)}{\underbrace{\sqrt{16 \omega_{p_1} \dots \omega_{p_4}}}_{\text{known}}} \end{aligned}$$

Goal: Calculate M .

$$M = \sum_{i,j} \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^4} \frac{d^4 p}{(2\pi)^4} \cdot \frac{1}{(-p^2 + m^2 - i\epsilon)(-(p_1 + p_2 - p)^2 + m^2 - i\epsilon)}$$

$$\tilde{M}_{12} = \lim_{\epsilon \rightarrow 0^+} \int_{\mathbb{R}^4} \frac{d^4 p}{(2\pi)^4} \cdot \frac{\underbrace{\Theta(p)}_{\text{OCP}}}{(-p^2 + m^2 - i\epsilon)(-(p_1 + p_2 - p)^2 + m^2 - i\epsilon)}$$

Since M diverges to ∞ as $\|p\| \rightarrow \infty$,
 $\Theta(p) = 1$ if $\|p\| < R$ and decay fast outside. } (*)
 $\Rightarrow \tilde{M} < \infty$. } THE condition
 ↗ cutoff fn.

But, we don't know Θ .

However, we can calculate it without knowing Θ
 if we know the value for some $(P_1^*, P_2^*, P_3^*, P_4^*)$
 by observation.

$$\langle P_3^*, P_4^* | S | P_1^*, P_2^* \rangle = A_* \frac{-P(2\pi)^* \delta^{(4)}(P_3^* + P_4^* - P_1^* - P_2^*)}{\sqrt{16\omega_{P_1^*} \dots \omega_{P_4^*}}}$$

We assume $P_3^* + P_4^* - P_1^* - P_2^* = 0$ to handle the value A_* .

$$A_* = g - ig^2 \tilde{M}^* + \mathcal{O}(g^3) \rightarrow g = A_* + ig^2 \tilde{M}^* + \mathcal{O}(g^3)$$

$$g^2 = A_*^2 + \mathcal{O}(g^3)$$

$$A = g - ig^2 \tilde{M} + \mathcal{O}(g^3)$$

$$= A_* + iA_*^2 \tilde{M}^* - iA_*^2 \tilde{M} + \mathcal{O}(g^3)$$

$$= A_* + iA_*^2 (\tilde{M}^* - \tilde{M}) + \mathcal{O}(g^3).$$

If cutoff Λ satisfies THE condition,
 $L := \lim_{R \rightarrow \infty} (\tilde{M}^* - \tilde{M})$ exists.

$$\therefore A \cong A_* + iA_*^2 L + \mathcal{O}(g^3).$$

> Counterterm Method.

$$g = g_r + ig_r^2 \tilde{M}^*.$$

$$A = g - ig^2 \tilde{M} + \mathcal{O}(g^3) = g_r + ig_r^2 \tilde{M}^* - i(g_r + ig_r^2 \tilde{M}^*)^2 \tilde{M} + \mathcal{O}(g^3)$$

$$= g_r - ig_r^2 (\tilde{M} - \tilde{M}^*) + \mathcal{O}(g^3).$$

$\downarrow$
 conv as $R \rightarrow \infty$

One-loop) $A = g - ig^2 M + O(g^3)$

Two-loop) $A = g - ig^2 M_2 + g^3 M_3 + O(g^4)$

→ Much much more complicated.

What we have done is perturbative renormalization. It becomes harder as order gets higher. If it can be done for all order, then the theory is called "perturbatively renormalizable".

21. One-loop Renormalization in ϕ^4 Theory.

We want to calculate

$$U(w, \theta) := \lim_{\varepsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \frac{\theta(p)}{(-p^2 + m^2 - i\varepsilon)(-(w-p)^2 + m^2 - i\varepsilon)}$$

where $w = p_1 + p_2$. & θ satisfies THE conditions.

One-loop renormalization Thm

For any $w, w' \in \mathbb{R}^{1,3}$,

$\lim_{\Lambda \rightarrow \infty} (U(w, \theta) - U(w', \theta))$ exists

and depends only on (w, w') . θ satisfies THE cond.

Lemma (Feynman Parameter) → no hole.

Supp $A, B \in \mathbb{C}$ s.t. the line segment btw A & B does not pass the origin. Then,

$$\frac{1}{AB} = \int_0^1 \frac{du}{(Au + B(1-u))^2} \quad \text{"Feynman parameter"}$$

(pf) $\frac{d}{du} \left[\frac{1}{B-A} \cdot \frac{1}{Au + B(1-u)} \right]_{u=0}^{u=1} = \frac{-\cancel{A+B}}{(Au + B(1-u))^2}$

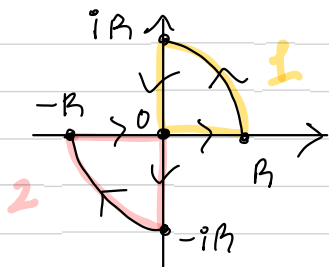
$$= \frac{1}{B-A} \left(\frac{1}{A} - \frac{1}{B} \right) = \frac{1}{AB}. \quad \square$$

Lemma (Wick Rotation)

Supp $f: \mathbb{C} \rightarrow \mathbb{C}$ is holomorphic & which has no poles in the region $\{x+iy : xy \geq 0\}$. Also, supp. $|f(z)| \leq C|z|^{1-\epsilon}$ for some $C < \infty, \epsilon > 0$.

Then, $\int_{-\infty}^{\infty} dx f(x) = i \int_{-\infty}^{\infty} dx f(ix).$

(pf)



$\square$

→ guarantees the integral is well-defined.

Cor For any $\alpha > 0, E \in \mathbb{R}, \epsilon > 0$,

$$\int_{-\infty}^{\infty} dx \frac{1}{\underbrace{(-\alpha x^2 + E - i\epsilon)^2}_{f(z)}} = i \int_{-\infty}^{\infty} dx \frac{1}{(\alpha x^2 + E - i\epsilon)^2}.$$

//
 $f(z)$

$$p = (p^0, p^1, p^2, p^3)$$

One-loop renormalization Thm

For any $w, w' \in \mathbb{R}^{4,3}$,

$\lim_{\Lambda \rightarrow \infty} (U(w, \theta) - U(w', \theta))$ exists
and depends only on (w, w') . θ satisfies (*).

$$\begin{aligned} \langle pA \rangle U(w, \theta) &= \lim_{\epsilon \rightarrow 0^+} \int \frac{d^4 p}{(2\pi)^4} \cdot \frac{\theta(p)}{(-p^2 + m^2 - i\epsilon)(-(w-p)^2 + m^2 - i\epsilon)} \\ &\stackrel{F.P.}{=} \lim_{\epsilon \rightarrow 0^+} \int_0^1 du \int \frac{d^4 p}{(2\pi)^4} \cdot \frac{\theta(p)}{(-(w-p)^2 u - p^2(1-u) + m^2 - i\epsilon)^2} \end{aligned}$$

Change of variable ($p \rightarrow p + uw$)

$$= \lim_{\epsilon \rightarrow 0^+} \int_0^1 du \int \frac{d^4 p}{(2\pi)^4} \frac{\theta(p + uw)}{(-p^2 - u(1-u)w^2 + m^2 - i\epsilon)^2}$$

$$\begin{aligned} &\int_{-\infty}^{\infty} \frac{dp^0}{2\pi} \cdot \frac{1}{(+(p^0)^2 + |p^2 - u(1-u)w^2 + m^2 - i\epsilon|^2)^2} \\ &= i \int_{-\infty}^{\infty} \frac{dp^0}{2\pi} \cdot \frac{1}{(\underbrace{\|p\|_E^2}_{\text{Euclidean norm}} - u(1-u)w^2 + m^2 - i\epsilon)^2} \end{aligned}$$

$$\begin{aligned} \therefore U(w, \theta) &= \lim_{\epsilon \rightarrow 0^+} \int_0^1 du \int \frac{d^4 p}{(2\pi)^4} \frac{\theta(p + uw)}{(\|p\|_E^2 - u(1-u)w^2 + m^2 - i\epsilon)^2} \\ &= \underbrace{U_1}_{\text{over } \|p\|_E \leq R_{1/2}} + \underbrace{U_2}_{\text{over } \|p\|_E > R_{1/2}}. \end{aligned}$$

Lemma $R \gg 0$. $u_1(w, \theta) = \frac{i}{(2\pi)^4} \int_0^1 du A(u)$

where $A(u) = \frac{1}{2} \left[\log \left(\frac{(R/2)^2 - u(1-u)w^2 + m^2}{(R/2)^2} \right) - \log(-u(1-u)w^2 + m^2) \right]$

where $\log(-x) = \log x - i\pi$ for $x > 0$.

(pf) If $R \gg 0$, $\theta(p+u) = 1$ for $\forall \|p\| \leq R/2, \forall u \in [0, 1]$.

Let $C = -u(1-u)w^2 + m^2 - i\epsilon$.

Using Polar coordinates,

$$u_1(w, \theta) = i \lim_{\epsilon \rightarrow 0^+} \int_0^1 du \int_{\|p\| \leq R/2} \frac{d^4 p}{(2\pi)^4} \frac{1}{(\|p\|^2 + C)^2}$$

$$= \lim_{\epsilon \rightarrow 0^+} \frac{i}{16\pi^4} \int_0^1 du \int_0^{R/2} \frac{r^3 dr}{(r^2 + C)^2}$$

$$\text{So, } \int_0^{R/2} \frac{r^3 dr}{(r^2 + C)^2} = \int_0^{R/2} \frac{r dr}{r^2 + C} - C \int_0^{R/2} \frac{r dr}{(r^2 + C)^2}$$

$$= \frac{1}{2} (\log((R/2)^2 + C) - \log C) + \frac{C}{2} \left(\frac{1}{(R/2)^2 + C} - \frac{1}{C} \right)$$

$$= \frac{1}{2} \left(\log((R/2)^2 + C) - \log C - \frac{(R/2)^2}{(R/2)^2 + C} \right)$$

Send $\epsilon \rightarrow 0^+$. $\square$

$$\therefore \lim_{R \rightarrow \infty} u_1(w, \theta) - u_1(w', \theta) = L(w) - L(w')$$

where $L(w) = \frac{-i}{(2\pi)^4} \int_0^1 du \log(m^2 - u(1-u)w^2)$

Lemma $\lim_{R \rightarrow \infty} u_2(w, \theta) - u_2(w', \theta) = 0.$

$$\begin{aligned}
 \langle p, p \rangle \quad u_2(w, \theta) &= i \lim_{\epsilon \rightarrow 0^+} \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} \frac{\theta(p+uw)}{(\|p\|^2 - u(1-u)w^2 + m^2 - \epsilon)^2} \\
 &= i \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} \frac{\theta(p+uw)}{(\|p\|^2 - u(1-u)w^2 + m^2)^2} \\
 &\stackrel{p \rightarrow p-uw \leftarrow}{=} i \int_0^1 du \int_{\|p-uw\| \geq R/2} \frac{d^4 p}{(2\pi)^4} \frac{\theta(p)}{(\|p-uw\|^2 - u(1-u)w^2 + m^2)^2} \\
 &= \underbrace{i \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} \frac{\theta(p)}{(\|p\|^2 - u(1-u)w^2 + m^2)^2}}_{u_3(w, \theta)} + \underbrace{\text{rest}}_{u_4(w, \theta)}
 \end{aligned}$$

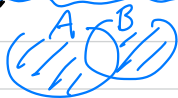
$$\begin{aligned}
 |u_3(w, \theta) - u_3(w', \theta)| &\leq \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} |\theta(p) F(p, u, w, w')| \\
 &\leq \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} |F(p, u, w, w')|
 \end{aligned}$$

where $F(p, u, w, w') = \frac{1}{(\|p\|^2 - u(1-u)w^2 + m^2)^2 (\|p\|^2 - u(1-u)w'^2 + m^2)^2}$

$\exists R_0, C > 0$ s.t. $\|p\| \geq R_0 \Rightarrow |F(p, u, w, w')| \leq C \|p\|^{-5}$

$\therefore \lim_{R \rightarrow \infty} |u_3(w, \theta) - u_3(w', \theta)| \leq \lim_{R \rightarrow \infty} \int_0^1 du \int_{\|p\| \geq R/2} \frac{d^4 p}{(2\pi)^4} C \|p\|^{-5} = 0.$

Let $\Delta(w, R)$ be symm difference of $\{p : \|p\| \geq R/2\}$ & $\{p : \|p-uw\| \geq R/2\}$.



Since $0 \leq \theta \leq 1$,

$$|u_4(w, \theta)| \leq \int_0^1 du \int_{\Delta(w, R)} \frac{d^4 p}{(2\pi)^4} \frac{1}{(\|p-uw\|^2 - u(1-u)w^2 + m^2)^2}.$$

Since $\text{Vol}(\Delta(w, R)) = \mathcal{O}(R^3)$ & $\curvearrowright$,

$|u_4(w, \theta)| = \mathcal{O}(R^{-1})$ as $R \rightarrow \infty$.

$$p = (p^0, p^1, p^2, p^3)$$

23. The Model for Free Photons.

Goal: Construct a model for free photon.

particle with mass m , velocity v

$$\rightarrow \text{four-momentum } p = \left(\frac{m}{\sqrt{1-v^2}}, \frac{mv}{\sqrt{1-v^2}} \right).$$

Photon: $m=0$, $|v|=1$.

$$p^2 = m^2 \Rightarrow p \in X_m.$$

Minkowski I.P.
relativistic energy

$$(p^0)^2 = (p^1)^2 + (p^2)^2 + (p^3)^2$$

Consider $X_0 = \{p : p^2 = 0, p^0 \geq 0\}$.

Actually, no point mass.

Photons have four-momentum in X_0 .

From Lecture 10, $\exists$ Lorentz invariant measure $\underline{\lambda}_0$.

Photon has spin 1. $\rightarrow L^2(X_0, d\lambda_0)$: not proper.

> Gupta-Bleuler Quantization.

Set $\mathcal{H} = L^2(X_0, d\lambda_0, \mathbb{C}^4)$, a vector space.

$\gamma \in \mathcal{H} \Leftrightarrow \gamma: X_0 \rightarrow \mathbb{C}^4$ and $\int |\gamma|^2 d\lambda_0 < \infty$.

For $w, z \in \mathbb{C}^4$, define the Minkowski inner product

$$(w, z) = (w^0)^* z^0 - (w^1)^* z^1 - (w^2)^* z^2 - (w^3)^* z^3.$$

For $\xi, \gamma \in \mathcal{H}$, define

$$(\xi, \gamma) = - \int d\lambda_0(p) (\xi(p), \gamma(p)). \dots (*)$$

$\Rightarrow (\xi, \gamma) = \overline{(\gamma, \xi)}$ sesquilinear form on $\mathcal{H}$.

But not positive-definite.

Define $\mathcal{H}' = \{ \xi \in \mathcal{H} : (p, \xi(p)) = 0 \text{ for } \lambda_0 \text{ a.e. } p \}$.

Lem The inner product (x) is positive-semidefinite in $\mathcal{H}'$.

$\langle p | \rangle$ Let $\xi \in \mathcal{H}'$. Choose $p \neq 0$ s.t. $(p, \xi(p)) = 0$.

$$p = (p^0, \mathbf{p}) = (p^0, p^1, p^2, p^3). \quad \text{(relativistic)} \\ \text{Since } p \in X_0, \quad \text{energy } \geq 0. \\ (p^0)^2 = (p^1)^2 + (p^2)^2 + (p^3)^2.$$

Since $p \neq 0$, $p^0 > 0$ and $\mathbf{p} \neq 0$.

Let $A \in SO(3)$ s.t. $A\mathbf{p} = (0, 0, a)$ with $a > 0$.

Since A is a rotation, $|a| = |\mathbf{p}| = p^0$.

Let $\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & A \end{pmatrix} \in SO^\uparrow(1, 3)$. : restricted Lorentz

Note $\Lambda p = (p^0, 0, 0, p^0)$. Let $x = \xi(p)$ and $z = \Lambda x$.

$$(p, x) = (\Lambda p, \Lambda x) = (\Lambda p, z) = p^0 (z^0 - z^3).$$

Since $(p, x) = (p, \xi(p)) = 0$ and $p^0 \geq 0$, $z^0 = z^3$.

$$\therefore (\xi(p), \xi(p)) = (z, z) = (z^0)^2 - (z^1)^2 - (z^2)^2 - (z^3)^2 \leq 0.$$

$$\therefore (\xi, \xi) \geq 0. \quad \square \quad \text{Lorentz Transformation.}$$

Define $\mathcal{H}_{\text{null}} := \{\xi \in \mathcal{H}' : (\xi, \xi) = 0\}$.

$\mathcal{H}_{\text{phys}} := \mathcal{H}' / \mathcal{H}_{\text{null}}$. $\rightarrow$ Hilbert sp under Minkowski inner product. (complete).

$\downarrow$
Hilbert space for a single photon.

For many photons, use the Bosonic Fock space of $\mathcal{H}_{\text{phys}}$.